

CALAPAN, Philippines

WMO: 984310

Lat: 13.4147N

Lon: 121.1868E

Elev: 41

StdP: 100.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	23.2	19.7	14.4	24.2	20.3	15.1	24.5	6.1	25.4	5.4	25.6	1.4	90	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.9	33.0	28.0	32.4	27.7	31.9	27.5	28.8	31.9	28.3	31.5	28.0	31.2	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.9	24.2	31.2	27.5	23.5	30.9	27.1	22.9	30.6	94.0	32.1	91.8	31.7	90.1	31.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.7	4.0	20.6	34.8	2.3	1.7	18.9	36.0	17.6	37.1	16.3	38.0	14.6	39.3		
			19.3	29.6	1.7	0.6	18.1	30.0	17.1	30.4	16.1	30.7	14.9	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	25.9	26.3	27.1	28.4	28.8	28.3	27.8	28.0	27.8	27.5	27.2	26.3
	DBStd	1.35	0.95	0.96	0.95	1.01	0.99	1.03	1.11	1.17	1.01	0.97	0.96	1.07
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6371	492	456	531	552	583	550	551	557	533	543	516	507
	CDD18.3	3329	233	223	273	302	325	300	292	299	283	285	266	248
	CDH23.3	33738	1704	1796	2612	3451	3887	3431	3081	3238	2974	2923	2585	2058
Wind	WSAvg	1.5	1.7	1.7	1.8	1.7	1.3	1.2	1.2	1.2	1.3	1.3	1.5	1.8
	PrecAvg	1885	87	47	51	80	176	173	208	185	180	264	240	182
	PrecMax	2547	215	192	161	286	468	338	313	326	487	547	976	409
	PrecMin	1036	24	11	0	12	31	24	22	26	37	106	43	53
	PrecStd	411	54	48	41	72	124	87	83	90	124	133	214	102
	DB	29.8	30.6	31.5	33.0	34.0	33.4	32.8	33.2	32.8	32.2	31.4	30.2	
	MCWB	25.9	26.2	26.7	27.6	28.7	28.3	28.2	28.2	28.2	27.7	27.6	27.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.0	29.7	30.6	32.2	33.0	32.6	31.9	32.2	31.7	31.3	30.5	29.6	
	2% MCWB	25.7	25.7	26.3	27.1	27.9	28.0	27.7	27.8	27.7	27.3	27.1	26.5	
	5% DB	28.4	29.1	30.1	31.6	32.3	31.9	31.2	31.4	31.1	30.7	29.9	28.9	
	5% MCWB	25.3	25.4	26.0	26.9	27.6	27.8	27.4	27.4	27.4	27.1	26.8	26.2	
	10% DB	27.9	28.5	29.5	30.9	31.7	31.2	30.5	30.7	30.5	30.0	29.3	28.3	
	10% MCWB	25.0	25.2	25.7	26.6	27.4	27.5	27.1	27.1	27.1	26.8	26.5	25.8	
	WB	26.7	27.0	27.6	28.6	29.2	29.2	28.9	29.1	29.1	28.5	28.5	27.7	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.9	29.7	30.5	31.9	32.9	32.3	31.8	32.0	31.9	31.1	30.7	29.6	
	2% WB	26.1	26.3	27.0	27.9	28.6	28.5	28.1	28.2	28.2	27.7	27.6	26.9	
	2% MCDB	28.3	29.0	30.0	31.4	32.2	31.8	31.1	31.4	31.1	30.5	29.9	29.0	
	5% WB	25.7	25.9	26.5	27.4	28.1	28.0	27.6	27.7	27.7	27.3	27.1	26.4	
	5% MCDB	27.8	28.5	29.4	30.8	31.6	31.3	30.6	30.8	30.6	30.1	29.5	28.4	
	10% WB	25.3	25.4	26.0	27.0	27.6	27.6	27.2	27.2	27.2	26.9	26.7	26.0	
	10% MCDB	27.3	27.9	28.8	30.2	31.0	30.8	30.1	30.2	30.0	29.6	28.9	27.8	
Mean Daily Temperature Range	MDBR	3.8	4.4	4.8	5.3	5.9	5.7	5.3	5.3	5.4	5.2	4.4	3.7	
	5% DB	4.7	5.2	5.5	6.1	6.8	6.7	6.3	6.5	6.4	6.1	5.1	4.6	
	MCWBR	2.8	3.0	3.0	3.0	3.6	3.8	3.6	3.7	3.7	3.5	3.1	2.8	
	5% WB	4.4	4.9	5.3	5.9	6.5	6.4	6.0	6.1	6.1	5.8	5.0	4.4	
	MCWBR	2.9	3.2	3.2	3.4	3.8	3.9	3.8	3.8	3.9	3.6	3.2	2.9	
Clear-Sky Solar Irradiance	taub	0.431	0.423	0.444	0.455	0.452	0.445	0.442	0.448	0.446	0.468	0.443	0.450	
	taud	2.410	2.420	2.361	2.336	2.372	2.411	2.417	2.392	2.391	2.312	2.392	2.353	
	Ebn at Noon	868	891	876	859	847	844	849	854	861	837	849	838	
	Edh at Noon	116	118	128	131	124	118	118	122	123	130	117	120	
All-Sky Solar Radiation	RadAvg	4.37	5.20	5.86	6.47	5.82	5.24	4.74	4.68	4.81	4.66	4.42	3.92	
	RadStd	0.35	0.49	0.45	0.35	0.53	0.43	0.54	0.37	0.49	0.32	0.39	0.35	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.25	+0.39	N/A	+0.44	+0.36	+0.45	N/A	N/A	+158	+158

Nomenclature: See separate page